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# Searching Short-Term Rentals in Porto and Lisbon





Faculdade de Engenharia da Universidade do Porto

M.EIC003 – Processamento e Recuperação de Informação

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# The Problem

Choosing a place to stay in a city is difficult when all the available information is scattered throughout the whole internet. It makes the process a lot slower and tiring.

A search system that brings this information together and allows travellers to ask free-text questions (such as “quiet accommodations near historic sites”) presents the most practical solution for this problem.

In order to develop this system, multiple data sources were used.

# Data Sources

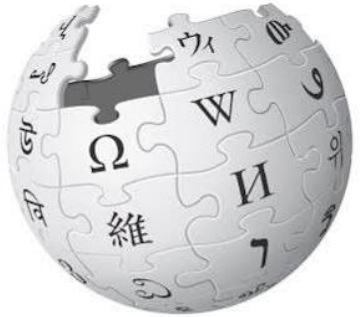
Inside Airbnb
Adding data to the debate

- Data source responsible for:
  - Listings
  - Reviews
  - Neighbourhoods



OpenStreetMap

- Data source responsible for:
  - Points of Interest

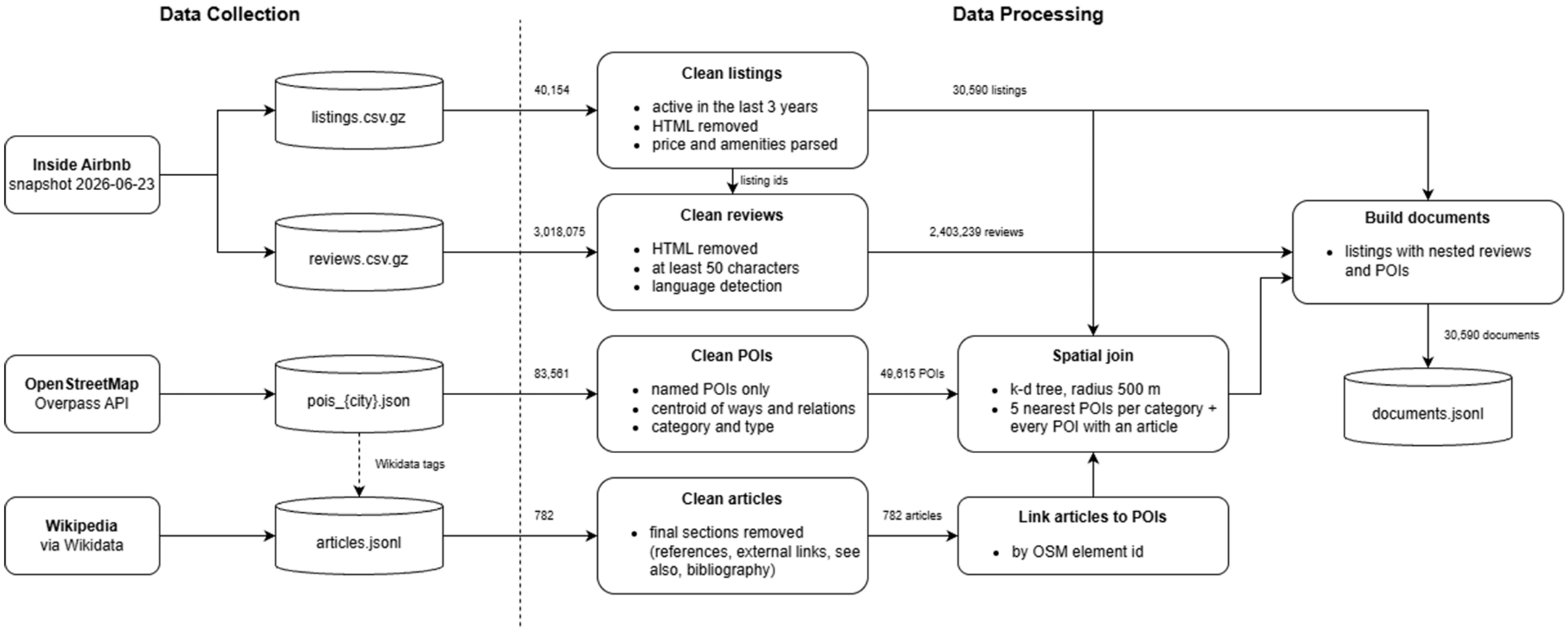


WIKIPEDIA
The Free Encyclopedia

- Data source responsible for:
  - Articles about Points of Interest

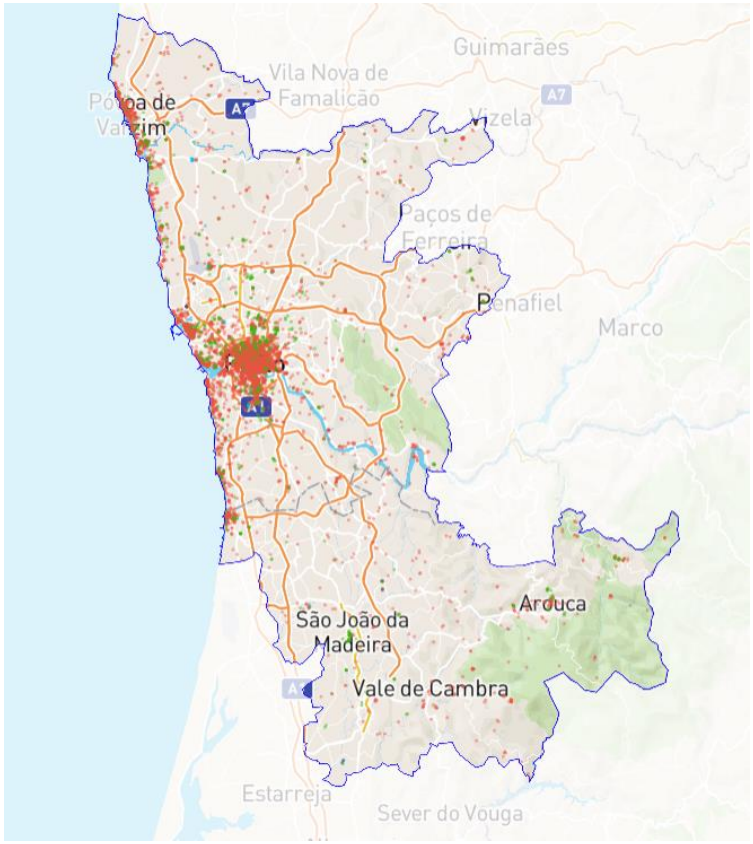
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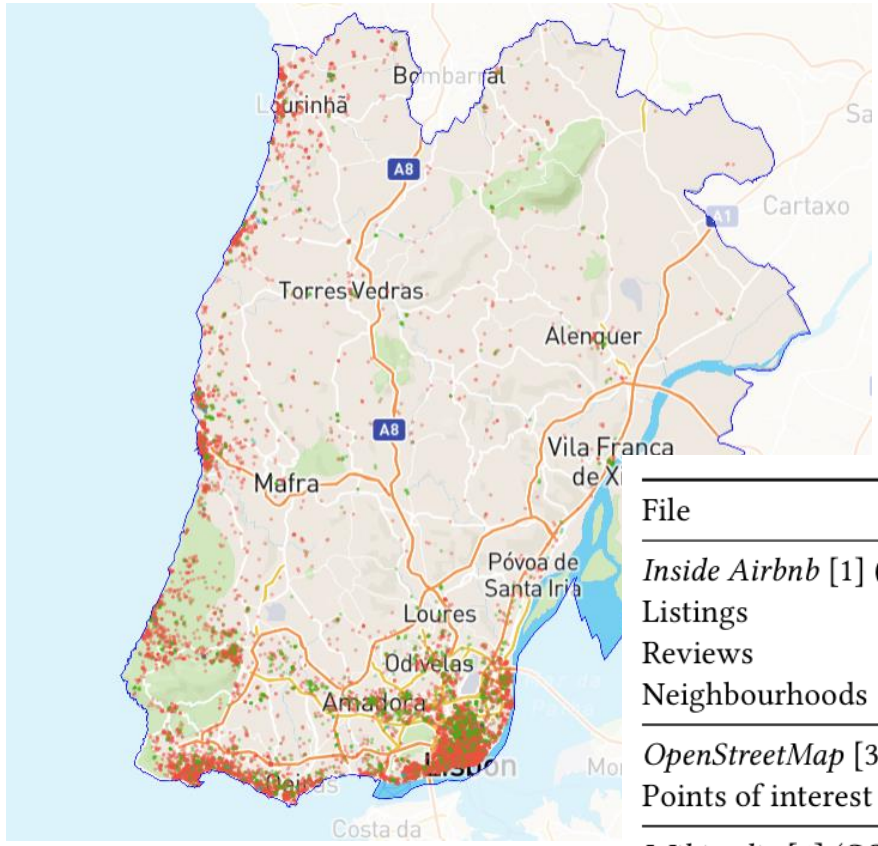
# Data Collection and Pipeline



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# Dataset

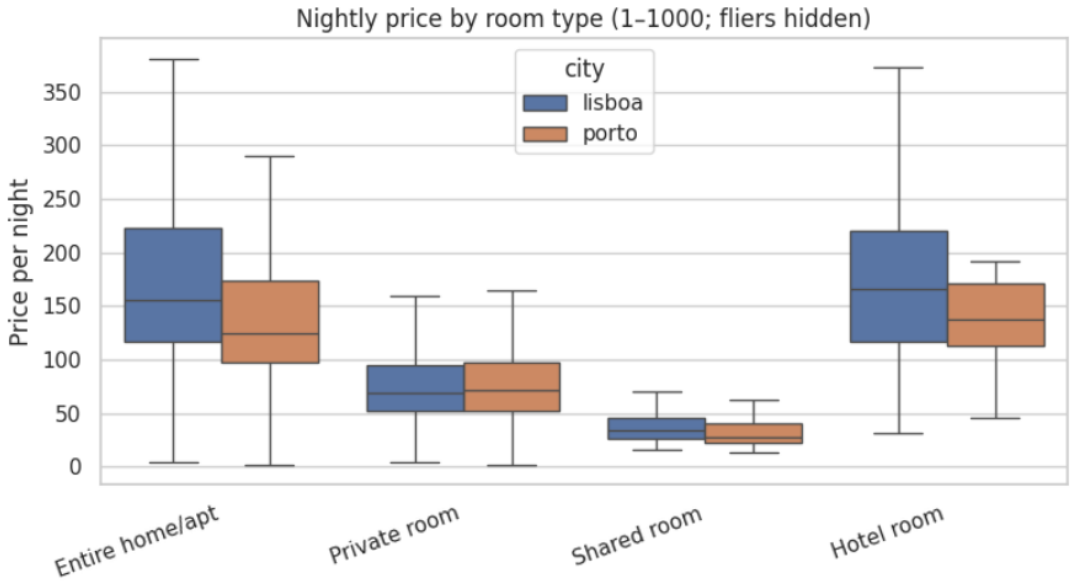


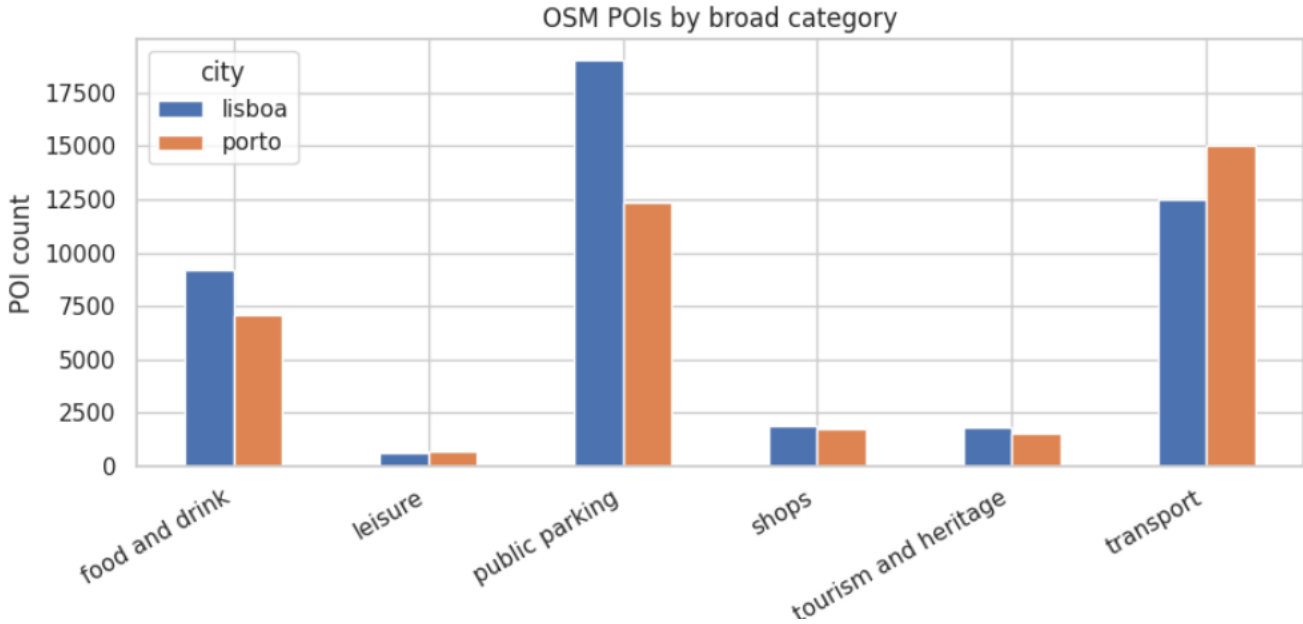




| File | Format | Porto | Lisbon | Size (MB) |
|---|---|---|---|---|
| *Inside Airbnb* [1] (CC BY 4.0) | | | | |
| Listings | CSV | 15,278 | 24,876 | 23.7 |
| Reviews | CSV | 1,131,143 | 1,886,932 | 395.2 |
| Neighbourhoods | GeoJSON | 177 | 173 | 7.2 |
| *OpenStreetMap* [3] (ODbL 1.0) | | | | |
| Points of interest | JSON | 38,390 | 45,162 | 26.2 |
| *Wikipedia* [6] (CC BY-SA 4.0) | | | | |
| Articles* | JSON Lines | 311 | 472 | 3.2 |

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# Preliminary Data Exploration



# Integration Challenge and Target Model

**Spatial Matching**
Airbnbs connected to POIs through coordinates, with a radius of 500m.

**Data Cleaning**
Airbnbs that are inactive and/or don’t have significant reviews (and reviews with less than 50 characters) are removed.

**Final Structure**
Each Airbnb becomes a document that contains its attributes, reviews and close POIs.